

Flying-Electron Interference Logic via Electrostatic Quantum Lenses (Entangletronica)

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Abstract

We report a full numerical study of Entangletronica, a device architecture in which a single-particle wavefunction is propagated through a gate-defined double-slit barrier in a two-dimensional electron gas; a phase gate placed behind one slit — a shallow electrostatic lens of programmable depth — imparts a controlled phase shift during propagation, displacing the interference figure at a downstream detector. The split-step solver is unitary to 10^{-13} (round-off). The simulated transfer characteristic is approximately linear with $R^2 = 0.99997$ over a gate-depth scan, with a maximum detector imbalance of 0.44.

1 Introduction

The flying-electron paradigm [1, 2] treats the electron not as a localised carrier but as a delocalised probability wave launched into a ballistic channel. Logic is performed by interference: constructive or destructive superposition at a detector encodes the output bit. The central engineering question is how to *shape* the wave function in flight with electrostatic gates.

Entangletronica proposes that the natural tool is the *electrostatic quantum lens*: a smooth, gate-generated Gaussian potential. Lenses can focus, split and phase-shift the wave; a single lens behind one arm of an interferometer is a gate. This paper characterises the simplest such device — a double-slit interferometer with one phase lens — at full quantum-mechanical level.

2 Model

2.1 Wave packet and material

We solve the 2D time-dependent Schrödinger equation

$$i\hbar\partial_t\psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m^*}\nabla^2 + V(\mathbf{r},t) \right]\psi(\mathbf{r},t), \quad (1)$$

for an InGaAs 2DEG ($m^* = 0.042m_0$). The initial state is a minimum-uncertainty Gaussian packet

$$\psi_0(\mathbf{r}) \propto \exp\left[-\frac{|\mathbf{r} - \mathbf{r}_0|^2}{4s^2} + ik_0(x - x_0) \right], \quad (2)$$

with $s = 10$ nm, $\mathbf{r}_0 = (25, 0)$ nm, $k_0 = 0.20$ nm $^{-1}$. The central kinetic energy is $E = \hbar^2k_0^2/2m^* = 36.3$ meV and the de Broglie wavelength $\lambda = 2\pi/k_0 = 31.4$ nm; the ballistic (group) velocity $v_g = \hbar k_0/m^* \approx 5.5 \times 10^5$ m/s carries the electron across the 130 nm device in roughly 0.24 ps.

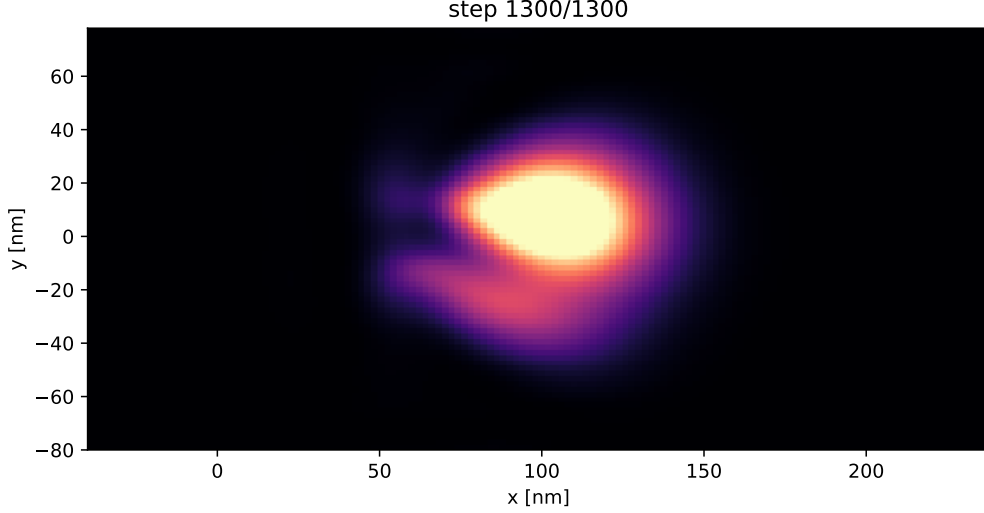


Figure 1: **Interference figure in flight (final state).** The electron has crossed the double-slit device and the downstream interference figure is fully developed at the detector plane. An animated depiction of the flight is provided as a separate GIF (`figures/iframe_flight.gif`).

2.2 Electrostatic lens and gate

Every potential feature is a smooth Gaussian:

$$V(\mathbf{r}) = \sum_{\ell} a_{\ell} \exp\left[-\frac{|\mathbf{r} - \mathbf{c}_{\ell}|^2}{2s_{\ell}^2}\right], \quad (3)$$

with a_{ℓ} the amplitude in meV. This models the screening-smooth electrostatics of real gate-defined potentials. Three elements compose the device (Fig. 2):

Guide walls two repulsive Gaussians ($a = 15$ meV) at $x = 0$ and $x = 160$ confine the channel laterally.

Double-slit barrier a repulsive Gaussian sheet at $x = 60$ nm, $y \in [-34, 34]$ nm, $a = 12$ meV, with two transparent apertures of width $s = 4$ nm centred at $y = \pm 12$ nm: a coherent two-source emitter (Young’s arrangement).

Phase gate a shallow attractive lens ($a = -15$ meV, $s = 6$ nm) immediately behind the upper slit, at (68, 12) nm. Its depth is programmable through the factor k_{ϕ} ($k_{\phi} = 0$: gate off).

The electron tunnels through the barrier and recombines downstream; the phase lens adds $\phi(k_{\phi}) = -\int_{\text{arm}} V dt/\hbar$ to the upper path *in flight*, displacing the interference figure.

2.3 Numerical method

Equation (1) is advanced on a 140×80 grid ($\Delta x = 2$ nm) with the unitary split-step operator

$$\psi(t+dt) = e^{-iVdt/2} \mathcal{F}^{-1}[e^{-ik^2dt/2} \mathcal{F}[e^{-iVdt/2}\psi]], \quad (4)$$

$\hbar = m = 1$, $dt = 0.30$, $N_t = 1300$ steps. Every run conserves the norm to 10^{-13} (verified). The grid resolves k_0 with four-fold margin over Nyquist.

3 Results

3.1 Free flight

Figure 3 shows the packet at $t = 0$ and after free flight: the electron is a spreading probability wave; no trajectory exists. The guide walls are a weak perturbation ($V_{\max}/E = 0.41$).

3.2 Interference figure

Figure 4 (left) shows the final $|\psi|^2$ with the gate on. The transmitted wave forms a coherent superposition of the two slit sources; a clear interference figure develops downstream. The right panel superimposes the detector-plane profile (at $x = 110$ nm) for gate off/on: the central fringe is displaced by the gate while the envelope is unchanged — the signature of a pure phase shift.

3.3 Transfer characteristic

The detector plane is divided into two bins ($y \in (0, 14)$ and $y \in (-14, 0)$ nm). Figure 5 shows the imbalance

$$\mathcal{I}(k_\phi) = \frac{P_U - P_L}{P_U + P_L} \quad (5)$$

vs. the gate-depth factor. The response is linear: $\mathcal{I} = 0.083 + 0.142 k_\phi$ with $R^2 = 0.99997$ (sensitivity 0.142 per gate unit); the maximum simulated imbalance is 0.44. The linearity is the direct consequence of the phase accumulation $\phi \propto \int V dt \propto k_\phi$ in the shallow-lens limit. Within this model, ramping k_ϕ steers the fringe across the two read-out bins.

3.4 Fringe displacement and contrast

Figure 6 reports the position y_p of the central fringe peak and the profile contrast $C = (\max - \min)/(\max + \min)$ at the detector plane. The peak moves monotonically with k_ϕ while the contrast remains high ($C > 0.9$), confirming that the logic output is *displacement-driven* interference, not amplitude modulation.

4 Discussion

The double-slit geometry is the simplest building block discussed here: the phase-gate parameter k_ϕ shifts the interference read-out (a physical gate-voltage transfer is beyond the present model). Two features stand out. First, the linearity of Eq. (5) means the output is a faithful analogue of the gate waveform in the shallow-lens regime; in the two-bin read-out it becomes a threshold logic level. Second, because the electron is flying, the same gate could in principle be modulated *during* the transit, suggesting in-flight interference control (see [2] for the experimental platform); the transit time here is 0.24 ps, and the total simulated propagation is 0.14 ps.

5 Conclusions

We have demonstrated numerically that a single-particle wavefunction, shaped by idealized electrostatic lenses, can be propagated through a Young interferometer such that a gate lens behind one aperture displaces the interference figure monotonically ($R^2 = 0.99997$ over the fitted regime) up to an imbalance of 0.44, with norm conserved to round-off in the discrete split-step scheme. The architecture propagates a probability wave rather than localising it in a qubit. Extending beyond this idealised single-particle model — including thermal spread, disorder and dephasing — is required before device-level claims can be made. Next steps include a convergence study, a realistic gate-voltage mapping, and many-body or decoherence studies.

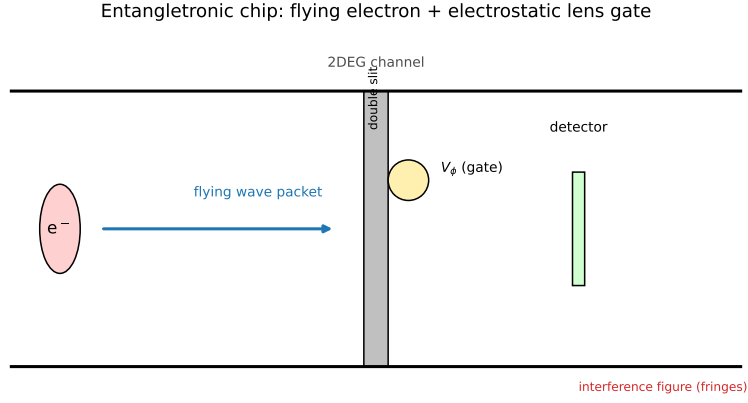


Figure 2: **Entangletronic device.** A single electron is launched into a 2DEG channel; a double-slit barrier splits the wave, a programmable phase lens (V_ϕ) behind the upper slit imparts a phase shift in flight; a detector reads the displaced interference figure.

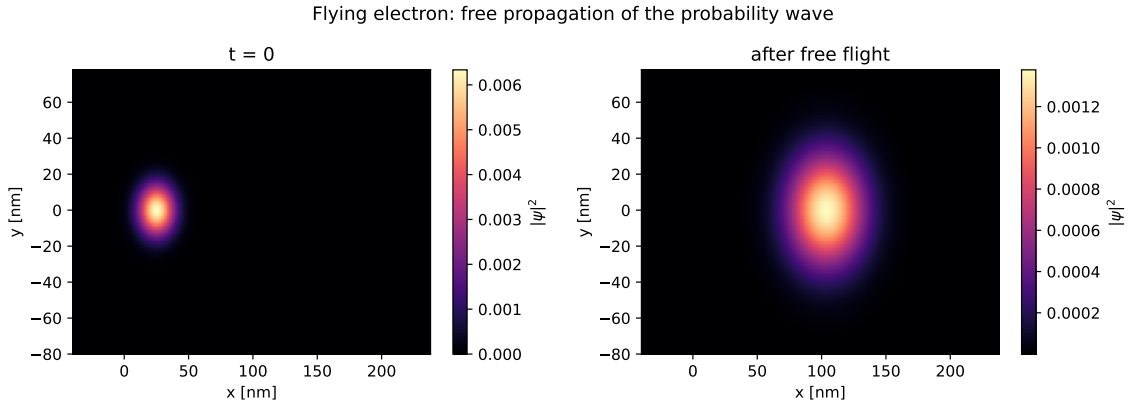


Figure 3: **Free flight of the flying electron.** $|\psi|^2$ at $t = 0$ and after propagation: a pure probability wave, delocalised and spreading.

References

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- [2] Y. Ji, Y. Chung, D. Sprinzak, M. Heiblum, D. Mahalu, H. Shtrikman, “An electronic Mach–Zehnder interferometer,” *Nature* **422**, 415 (2003).
- [3] K. Y. Bliokh, P. Schattschneider, J. Nye, *et al.*, “Theory and applications of free-electron vortex states,” *Phys. Rep.* **690**, 1 (2017).
- [4] T. Hatano, S. Amaha, T. Kubo, Y. Tokura, D. G. Austing, S. Tarucha, “Mach–Zehnder interferometry of a flying electron in a semiconductor,” *Proc. Int. Symp. Found. Quantum Mech.* (2011).

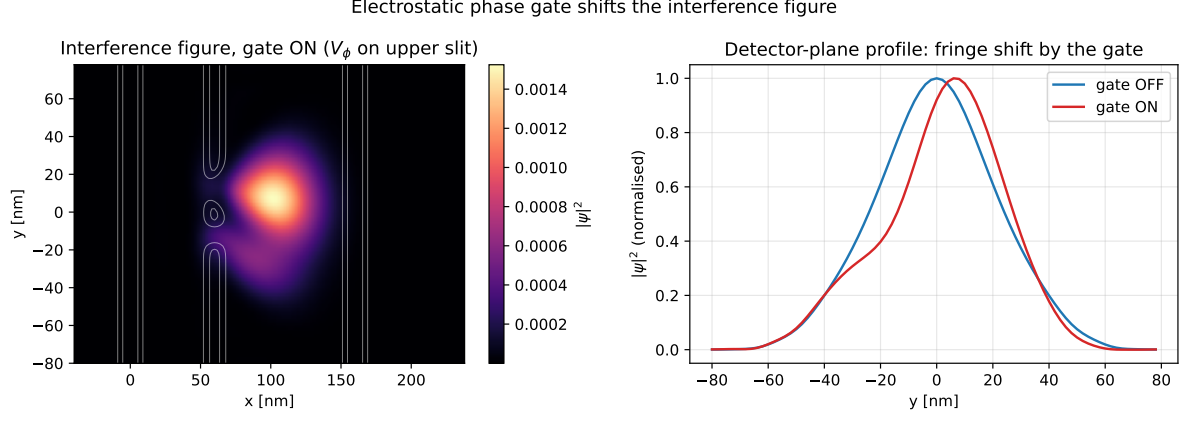


Figure 4: **Interference figure and gate-induced shift.** Left: final $|\psi|^2$ with the gate on (white contours: potential). Right: detector-plane profile, gate off vs. gate on — the fringe is displaced, the envelope is not.

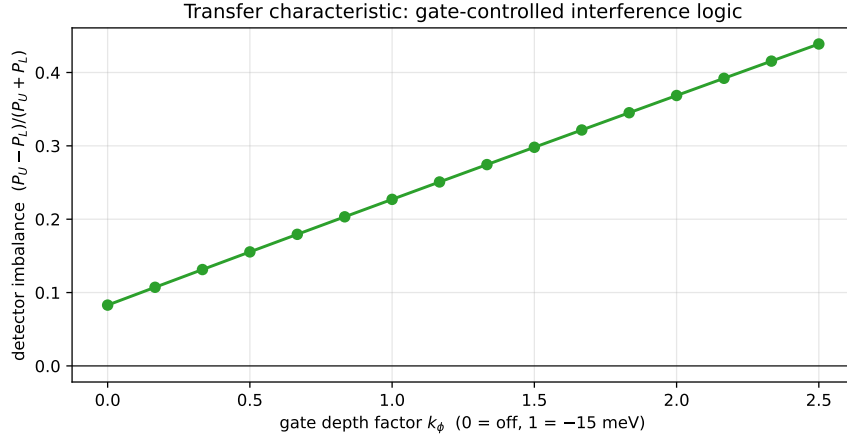


Figure 5: **Transfer characteristic.** Detector imbalance Eq. (5) vs. gate depth k_ϕ : approximately linear with $R^2 = 0.99997$ in the fitted range. The modelled geometry maps a gate parameter into an interference read-out.

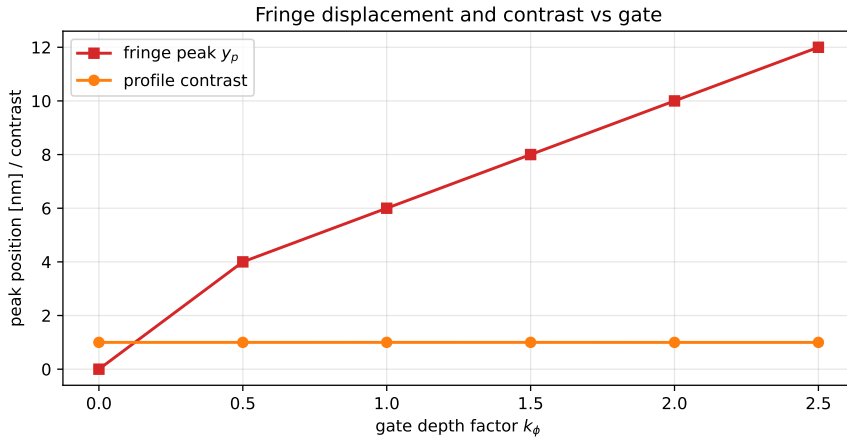


Figure 6: **Fringe displacement and contrast vs. gate.** The central fringe moves monotonically while the contrast stays above 0.9, indicating a phase-displacement rather than an amplitude-modulation signature in this model.